

Gabriel Ramirez-Garcia

Queens University
Universidad Nacional Autónoma de México

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Current Position

2022 – present

Postdoctoral Fellow in Neuroimaging

Project: Rehearsal therapy after stroke in non-human primates.
Project: Multimodal analysis in concussion football players.
Centre for Neurosciences Studies
Queen's University.
PI: DJ Cook, MD, PhD

2020 – 2021

Postdoctoral Fellow in Neuroimaging

Project: Neuronal bases of motor and cognitive impairment of polyglutamine disease patients by MRI.
Neuropsychology Lab, School of Medicine. Universidad Nacional Autónoma de México.
PI: Juan Fernandez-Ruiz, PhD.

Education

2015 – 2020

PhD in Biomedical Sciences.

Universidad Nacional Autónoma de México. School of Medicine.
Thesis: Longitudinal atrophy characterization of cortical and subcortical gray matter in Huntington's disease patients.
Subproject: Functional connectivity characterization in a MPTP parkinsonian macaque model.
Supervisor: Aurelio Campos Romo, MD, PhD.

2007 – 2013

BSc in Pharmacobiology Chemistry.

Benemerita Universidad Autónoma de Puebla. School of Chemistry.
Thesis: Nitrosative and cognitive effects of chronic L-DOPA administration in rats with an intra-nigral 6-OHDA lesion.
Supervisor: I Daniel Limón Pérez de León, PhD.

Academic stays

2017 – 2018

Centre for Neuroscience Studies at Queen's University, Kingston, Ontario Canada, Translational Stroke Research Lab, Department of Surgery, Division of Neurosurgery, Kingston General Hospital.

PI: DJ Cook, MD, PhD.

2014 – 2015

Neuropharmacology Lab, Pharmacy department, Benemerita Universidad Autonoma de Puebla. School of Chemistry.
PI: Daniel Limón Pérez de León, PhD.

Awards/Fellowships/Scholarships

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| 2025 | Renew of SNI appointment by the National System of Researchers in Mexico – Category I (660496). Mexico. |
| 2022 | SNI appointment by the National System of Researchers in Mexico – Category I (660496). Mexico. |
| 2022 | Mitacs Accelerate program postdoctoral grants in Canada 2020 CONACYT-Mexico fellowship 574022/403010 (CVU: 660496). Canada. |
| 2017 | International Brain Organization travel award. 2017 Housing Grant by Society for Neuroscience. US-Mexico. |
| 2015 | Latin American Training Program Award, Universidad Nacional Autonoma de Mexico, and the Society for Neuroscience. US-Mexico. |
| 2015 | CONACYT-Mexico fellowship 574022/403010 (CVU: 660496). |
| 2013 | Housing Grant by Society for Neuroscience. US-Mexico. |
| 2012 | Winner of the young researcher meeting (Chemistry – Biology category) by the CONCYTEP-CONACYT. Mexico. |
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Publications

1. F. Miranda-Casasola, **G. Ramirez-Garcia**, J Fernandez-Ruiz. (2026). Spatiotemporal indicators of dynamic gait stability using truncal accelerometry in healthy adults. Biomedical Signal Processing and Control. Biomedical Signal Processing and Control <https://doi.org/10.1016/j.bspc.2026.110964>
2. Rodriguez-Butler, Uriel, **Ramirez-Garcia, Gabriel**, Romero-Rebollar, Cesar, Estudillo-Guerra, Maria Anayali, Cancino, Valeria, Galvez, Víctor. Resting State-fMRI as a Complementary Tool to Explore Cognitive Impairment in Parkinson's Disease: A Systematic Review. Available at <http://dx.doi.org/10.2139/ssrn.5178880>. (peer review).
3. Sebastian D'Amario, Blaire Magee, Allen A. Champagne, Madison E. Wilson, Nicole S. Coverdale, **Gabriel Ramírez-García**, Douglas J Cook. Repeated subconcussive head impacts compromise white matter integrity: A longitudinal DTI and NODDI study in collegiate football players. bioRxiv 2025.05.14.654080; doi: <https://doi.org/10.1101/2025.05.14.654080>. (peer review).
4. Padron-Rivera, G., **Ramirez-Garcia, G.**, Chirino-Perez, A., Romero-Molina, A. O., Ochoa-Morales, A., Garcia-Gomar, M. G., Ramirez-Garcia, M. A., Rodriguez-Mendoza, O., Torres-Vences, D. L., Schüle, B., Pasaye-Alcaraz, E. H., Hernandez-Castillo, C. R., & Fernandez-Ruiz, J. (2026). Functional connectivity alterations in spinocerebellar ataxia type 10: insights from gray matter atrophy. *Brain imaging and behavior*, 20(1), 3. <https://doi.org/10.1007/s11682-026-01091-4>. **Cited by 1**
5. Romero-Molina AO, **Ramirez-Garcia G**, Chirino-Perez A, Padron-Rivera G, Hernandez- Castillo CR, Garcia-Gomar MG, et al. (2025) Cerebellar cognitive affective syndrome in patients with spinocerebellar ataxia type 10. PLoS ONE 20(3): e0319505. <https://doi.org/10.1371/journal.pone.0319505>. **Cited by 1**

6. Champagne, A. A., Coverdale, N. S., Skinner, C., Schwarz, B. A., Glikstein, R., Melkus, G., Murray, C. I., **Ramirez-Garcia, G.**, & Cook, D. J. (2025). Longitudinal analysis highlights structural changes in grey- and white-matter within military personnel exposed to blast. *Brain injury*, 39(6), 509–517. <https://doi.org/10.1080/02699052.2024.2446948>. **Cited by 1**
7. Romero-Molina, A.O., **Ramirez-Garcia, G.**, Chirino-Perez, A. et al. SARS-CoV-2's brain impact: revealing cortical and cerebellar differences via cluster analysis in COVID-19 recovered patients. *Neurol Sci* 45, 837–848 (2024). <https://doi.org/10.1007/s10072-023-07266-x>. **Cited by 2**
8. **Ramirez-Garcia, G.**, Escutia-Macedo, X., Cook, D. J., Moreno-Andrade, T., Villarreal-Garza, E., Campos-Coy, M., Elizondo-Riojas, G., Gongora-Rivera, F., Garza-Villarreal, E. A., & Fernandez-Ruiz, J. (2024). Consistent spatial lesion-symptom patterns: A comprehensive analysis using triangulation in lesion-symptom mapping in a cohort of stroke patients. *Magnetic resonance imaging*, 109, 286–293. <https://doi.org/10.1016/j.mri.2024.03.031>.
9. López-Ornelas, A., Escobedo-Avila, I., **Ramírez-García, G.**, Lara-Rodarte, R., Meléndez- Ramírez, C., Urrieta-Chávez, B., Barrios-García, T., Cáceres-Chávez, V. A., Flores-Ponce, X., Carmona, F., Reynoso, C. A., Aguilar, C., Kerik, N. E., Rocha, L., Verdugo-Díaz, L., Treviño, V., Bargas, J., Ramos-Mejía, V., Fernández-Ruiz, J., ... Velasco, I. (2023). Human Embryonic Stem Cell-Derived Immature Midbrain Dopaminergic Neurons Transplanted in Parkinsonian Monkeys. *Cells*, 12(23), 2738. <https://doi.org/10.3390/cells12232738>. **Cited by 10**
10. **Ramírez-García G.**, Galvez V, Diaz R., Campos-Romo A., Fernandez-Ruiz J. (2021). Montreal Cognitive Assessment (MoCA) performance in Huntington's disease patients correlates with cortical and caudate atrophy. *PeerJ. In Press*. **Cited by 6**
11. Flores-Vazquez JF., **Ramírez-García G.**, Marrufo-Meléndez O., Alcalá-Lozano R., Lietz MP., Rodríguez-Agudelo Y., Acosta-Castillo GI, Renken RJ., Aleman A., Enriquez-Geppert S., Sosa-Ortiz AL. (2021). Anosognosia in Amnesic Mild Cognitive Impairment is Related to Diminished Hippocampal Volume Comparable to Alzheimer's Disease Dementia: Preliminary MRI Findings. *Frontiers in Aging Neuroscience*. <https://doi.org/10.3389/fnagi.2021.739422>. **Cited by 2**
12. Chirino-Pérez, A., Vaca-Palomares I., Torres DL., Hernandez-Castillo, CR., Diaz, R., **Ramirez-Garcia, G.**, Fernandez-Ruiz J. (2021). Cognitive Impairments in Spinocerebellar Ataxia Type 10 and their Relation to Cortical Thickness. *Mov Disord*. <https://doi.org/10.1002/mds.28728>. **Cited by 18**
13. Chirino-Pérez, A., Marrufo-Meléndez OA., Muñoz-López, JI., Hernandez-Castillo, CR., **Ramirez-Garcia, G.**, Diaz, R., Nuñez-Orozco L., Fernandez-Ruiz J. (2021). Mapping the Cerebellar Cognitive Affective Syndrome in Patients with Chronic Cerebellar Strokes. *The Cerebellum*. <https://doi.org/10.1007/s12311-021-01290-3>. **Cited by 37**
14. Calderon-Villalon J, **Ramirez-Garcia G**, Fernandez-Ruiz J, Sangri-Gil F, Campos-Romo A, Galvez V. (2021). Planning deficits in Huntington's disease: A brain structural correlation by voxel-based morphometry. *PLOS ONE* 16(3): e0249144. <https://doi.org/10.1371/journal.pone.0249144>. **Cited by 5**
15. Contreras, A., **Ramirez-Garcia, G.**, Chirino, A. et al. (2020). Longitudinal Analysis of the Relation Between Clinical Impairment and Gray Matter Degeneration in Spinocerebellar Ataxia Type 7 Patients. *The Cerebellum*. <https://doi.org/10.1007/s12311-020-01205-8>. **Cited by 6**
16. **Ramirez-Garcia G**, Galvez V, Diaz R, Bayliss L, Fernandez-Ruiz J, Campos-Romo A. (2019). Longitudinal atrophy characterization of cortical and subcortical gray matter in Huntington's disease patients. *Eur J Neurosci*. 00: 1–17.

- <https://doi.org/10.1111/ejn.14617>. Cited by 15
17. Parra I, Martínez I, **Ramírez-García G**, Tizabi Y, Mendieta L. (2019) Differential Effects of LPS and 6-OHDA on Microglia's Morphology in Rats: Implications for Inflammatory Model of Parkinson's Disease. *Neurotox Res.* 37(1):1-11. <https://doi.org/10.1007/s12640-019-00104-z>. Cited by 35
 18. Palafox-Sanchez V, Sosti V, **Ramírez-García G**, Kulisevsky J, Aguilera J, Limón ID. (2019). Differential Expression of Striatal Δ FosB mRNA and FosB mRNA After Different Levodopa Treatment Regimens in a Rat Model of Parkinson's Disease. *Neurotox Res.* 35(3):563-574. <https://doi.org/10.1007/s12640-018-9993-0>. Cited by 6
 19. **Ramírez-García G**, Harrison KA, Fernandez-Ruiz J, Nashed JY, Cook DJ. (2019). Stroke Longitudinal Volumetric Measures Correlate with the Behavioral Score in Non-Human Primates. *Neuroscience.* 15(397):41-55. <https://doi.org/10.1016/j.neuroscience.2018.11.026>. Cited by 8
 20. Galvez V, **Ramírez-García G**, Hernandez-Castillo CR, Bayliss L, Díaz R, Lopez-Titla MM, Campos-Romo A, Fernandez-Ruiz J. (2017). Extrastriatal degeneration correlates with deficits in the motor domain subscales of the UHDRS. *J Neurol Sci.* 15385:22-29. <https://doi.org/10.1016/j.jns.2017.11.040>. Cited by 8
 21. Palafox-Sánchez V, Mendieta L, **Ramírez-García G**, Candalija A, Aguilera J, Limón ID. (2016). Effect of the C-terminal domain of the heavy chain of tetanus toxin on dyskinesia caused by levodopa in 6-hydroxydopamine-lesioned rats. *Pharmacol Biochem Behav.* 145:33-44. <https://doi.org/10.1016/j.pbb.2016.04.001>. Cited by 7
 22. **Ramírez-García G**, Palafox-Sánchez V, Limón ID. (2015). Nitrosative and cognitive effects of chronic L-DOPA administration in rats with intra-nigral 6-OHDA lesion. *Neuroscience.* 290: 492-508. <https://doi.org/10.1016/j.neuroscience.2015.01.047>. Cited by 28

Source of citing scores

<https://scholar.google.com/citations?user=95xkUm0AAAAJ&hl=en>

Teaching experience

2020 – present	Professor of the Principles of Cognitive Neurosciences course. Universidad Nacional Autonoma de México. School of Medicine. PhD program in Biomedical Sciences.
2020 – present	Professor of the Computational Models II – Magnetic resonance imaging analysis course. Universidad Nacional Autonoma de México. School of Medicine. Neuroscience program.
2020	Professor of Principles of Biomedical Instrumentation Universidad Nacional Autonoma de México. School of Engineering. Mater program in Biomedical Instrumentation.
2018 – present	Professor of the Neurophysiology course. Universidad Nacional Autonoma de México. School of Medicine. Neuroscience program.
2019	Professor of the Science of data in R course. Universidad Nacional Autonoma de México. School of Psychology. Psychology program
2019	Professor of the Principles of R programming course.

Work experience

2022 – present	Postdoctoral Fellow in Stroke Neuroimaging.
2020 – 2021	Postdoctoral Fellow in Neuroimaging in Motor diseases.
2013 – 2014	Laboratory supervisor at NicroBolta company, Puebla city, México.

Academic and research skills

Research Project Design and Collaboration: Designed, coordinated, and executed clinical and preclinical neuroscience studies, including hypothesis generation, experimental design, longitudinal data acquisition, and multimodal data analysis. Contributed to collaborative grant writing and funding proposals focused on therapeutic interventions, neuroimaging biomarkers, and translational neuroscience in both animal models and human patient populations.

Laboratory Management: Managed day-to-day laboratory operations, including coordination of research personnel, experimental scheduling, and compliance with institutional safety and animal care protocols. Contributed to the optimization of laboratory workflows and neuroimaging data acquisition procedures in both preclinical and clinical environments.

Teaching in Neuroimaging and Neurosciences: Instructor for undergraduate- and graduate-level courses in neuroimaging, cognitive neuroscience, MRI data analysis, and computational neuroscience methods. Developed course materials and practical training sessions focused on neuroimaging analysis and quantitative neuroscience approaches.

Research Supervision: Mentored and supervised undergraduate, master's, and PhD students in experimental design, neuroimaging methodologies, statistical analysis, programming workflows, and scientific writing. Provided training in multimodal MRI processing, reproducible research practices, and translational neuroscience research methodologies.

Human Neuroimaging Studies: Extensive experience in the analysis of structural, functional, and diffusion MRI across stroke and neurodegenerative disease populations, including Huntington's disease, Parkinson's disease, spinocerebellar ataxias, and Alzheimer's disease. Expertise includes lesion and white-matter hyperintensity segmentation, volumetric and cortical thickness analysis, white-matter microstructure assessment, T1/T2 myelin-sensitive mapping, resting-state fMRI, functional connectivity, manifold and gradient analysis, ICA decomposition, graph theory and seed-based connectivity analyses, GLM implementation, DTI tractography, and fixel-based analysis. Experienced in integrating multimodal MRI metrics with behavioral and clinical outcome measures.

Multimodal MRI Analysis: Extensive experience processing and analyzing structural, functional and diffusion MRI data, including T1w, T2w, FLAIR, DWI/DTI, resting-state fMRI, and TOF angiography across both human and non-human primate datasets.

Computational and Programming Expertise: Advanced proficiency in Bash, Python, R, and MATLAB for neuroimaging processing, statistical analysis, machine learning, and pipeline development. Extensive experience working in Linux/Ubuntu and macOS

environments, including high-performance and command-line-based workflows. Skilled in multimodal neuroimaging software packages including FreeSurfer, FSL, AFNI, SPM, MRtrix3, ANTs, GRETNA, and VolBrain.

MRI System Operations: Experience operating Siemens 3T MRI systems for neuroimaging studies, including subject preparation, scan protocol implementation, quality assurance, and adherence to MRI safety procedures and research compliance standards.

Non-Human Primate Research: Extensive experience conducting behavioral assessment, motor training, experimental manipulation, MRI preparation, physiological monitoring, and multimodal MRI analysis in Cynomolgus and Macaca mulatta models of neurological disease and recovery.

Animal Surgery and Modeling: Assisted and performed stereotaxic surgical procedures in rodents and non-human primates for the induction of neurodegenerative disease models, intracerebral delivery of therapeutic agents, electrode implantation, and neuromodulation-related experimental interventions.

Molecular Biology Techniques: Acquired technical expertise in molecular and cellular neuroscience methodologies.

Activities and interests

- *Active reader:* Particularly interested in reading sci-fiction novels.
- *Amateur painter:* Especially interested in learning gradient colour technique on canvas.
- *Regular cyclist:* Usually cycling 30 km on average as a weekend recreational activity.

Languages

Spanish. Native language.
English. Certified by the Canadian English Language Proficiency Index Program (CELPIP), January 29, 2025.

References

References available upon request.